Gelard Mutwezi

+256 772 676 944 | geraldvaros@gmail.com | linkedin.com/in/gerald-varos | github.com/Varos14 | Kampala, Uganda

EDUCATION

Victoria University Aug. 2022 -- 2025
Bachelor of Information Technology *Kampala, Uganda*

- **Key Coursework:** Software Engineering (A), Database Management Systems (A), Industrial Training (A), GIS (A), Application Development & DB Design (B+), Network & Information Security (B+), IT Project Management (B+).

London College of St. Lawrence 2020 -- 2022
Uganda Advanced Certificate of Education (UACE) — A-Level *Maya, Uganda*

London College of St. Lawrence 2016 -- 2019
Uganda Certificate of Education (UCE) — O-Level *Maya, Uganda*

EXPERIENCE

Junior IT Specialist Intern Mar. 2025 -- May 2026
Equal Opportunities Commission (Kingdom Kampala) *Kampala, Uganda*

- Delivered comprehensive IT support across commission departments, diagnosing and resolving hardware, software, and peripheral issues for staff workstations.
- Maintained enterprise local area network (LAN) infrastructure, managed IP configurations, network switches, and ensured stable internet connectivity.
- Administered user account provisioning, system access control, and enforced data backup and IT security policies to safeguard sensitive organizational records.
- Performed routine preventative maintenance and troubleshooting on network printers, scanners, and workstations, significantly minimizing departmental downtime.

Full-Stack Developer & Software Engineer Jan. 2024 -- Present
Wampeewo Ntake SS Project *Kampala, Uganda*

- Architected and built a full-stack Competency Class Management System (CMS) managing user access, student activities, assignment submissions, skill tracking, and daily attendance records.
- Implemented secure RESTful APIs in **Node.js**, **Express**, and **TypeScript**, integrated with **MySQL** via **Sequelize ORM** including automated database migration scripts and robust connection handling.
- Strengthened API security and performance using **JWT** role-based authentication (RBAC), **Helmet** HTTP security headers, and express rate-limiting to mitigate unauthorized access and DDoS vulnerabilities.

IT Specialist / Industrial Training Intern May 2024 -- Aug. 2024
Industrial Training Placement (Victoria University) *Kampala, Uganda*

PROJECTS

Wampeewo Competency CMS | *React 18, TypeScript, Node.js, Express, MySQL, TailwindCSS, Zustand* 2024

- Engineered a modular single-page application (SPA) featuring dynamic role-based routing, responsive dashboards, and interactive timetables with conflict-free study session scheduling.
- Integrated a rich-text WYSIWYG editor for student class notes with persistent database storage and state management using **Zustand** and **React Hook Form**.
- Containerized database and backend services using **Docker Compose** for streamlined cross-environment deployment.

Undergraduate Capstone Information System | *Systems Analysis, Relational DB, Security* 2024 -- 2025

- Conducted comprehensive system analysis and design (ER diagrams, data flow diagrams, architectural blueprints) to solve real-world organizational data management challenges.
- Structured 3NF relational database schemas with strict foreign-key integrity constraints, optimized indexing, and transaction-safe operations.

Geographical Information Systems (GIS) Spatial Mapping | *Spatial Data, Mapping APIs, Python* 2024

- Developed spatial analysis and mapping models for geographic coordinate data, achieving an **A Grade (85%)** in academic evaluation.
- Processed vector and raster layers to visualize regional demographic and infrastructure data points.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript (ES6+), SQL (MySQL, PostgreSQL, SQLite), Python, HTML5, CSS3
Frontend: React.js, Vite, TailwindCSS, Zustand, React Router, React Hook Form, Responsive Web Design
Backend & APIs: Node.js, Express.js, RESTful APIs, Sequelize ORM, PDFKit, JWT Authentication
Databases & DevOps: MySQL, SQLite, Docker, Docker Compose, Git, GitHub Actions, Linux / Bash
IT & Systems: Database Administration, Network Configuration (LAN/TCP/IP), Computer Hardware/OS Maintenance